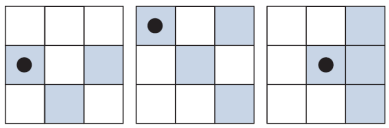
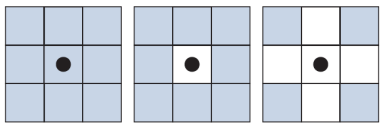
In the 4th version.

9.1 Find the reflection, $\hat{B}$, of each of the following structuring elements. The dot indicates the origin of the SE.



(a)* **(b)** **(c)**

9.2 Sketch the result of eroding Fig. 9.3(a) with each of the following structuring elements.



(a)* **(b)** **(c)**

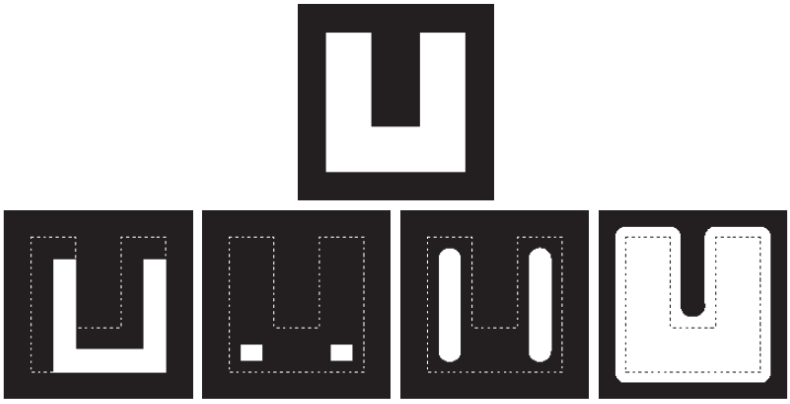
9.8 With reference to the image at the top of the figure shown below, answer the following:

(a)* Give the structuring element and morphological operation(s) that produced image (a). Show the origin of the structuring element. The dashed lines denote the boundary of the original object and are shown for reference; they are not part of the result. (The white elements are foreground pixels.)

(b) Repeat part (a) for the output shown in image (b).

(c)* Repeat part (a) for the output shown in image (c).

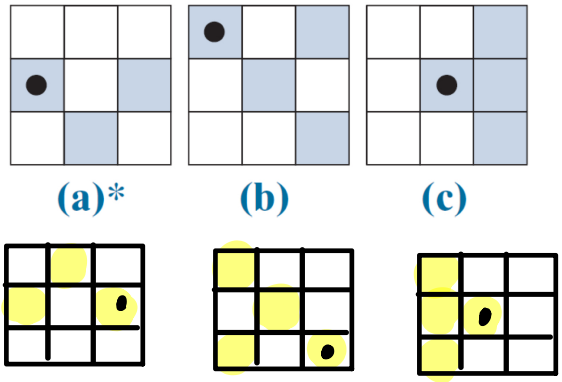
(d) Repeat part (a) for the solution shown in figure (d). Note that in image (d) all corners are rounded.



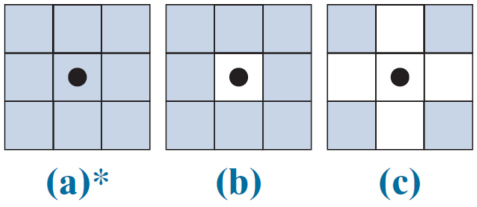
(a)* (b) (c)* (d)

DIP HW4

9.1 Find the reflection, $\hat{B}$, of each of the following structuring elements. The dot indicates the origin of the SE.

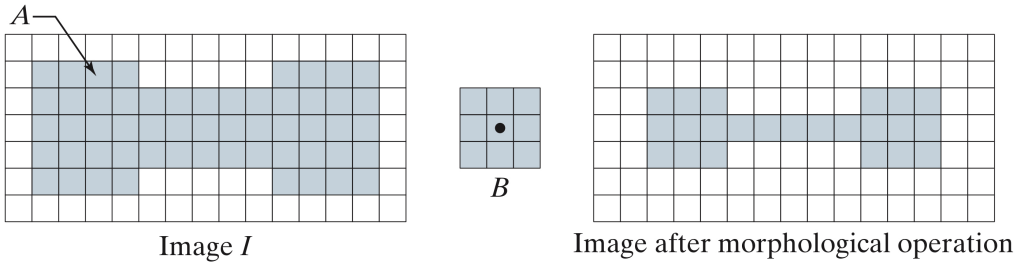


9.2 Sketch the result of eroding Fig. 9.3(a) with each of the following structuring elements.

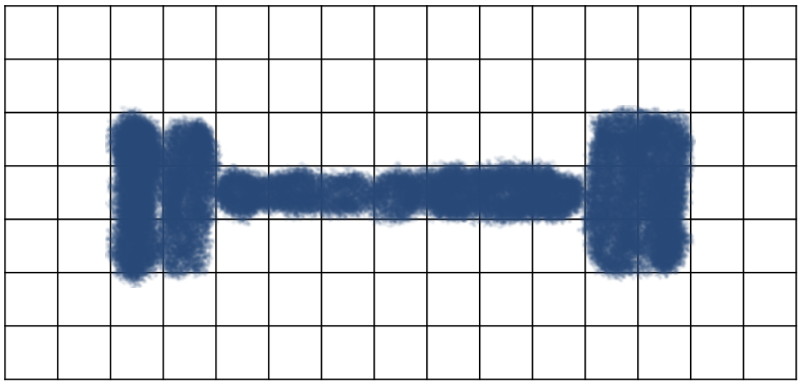


a b c

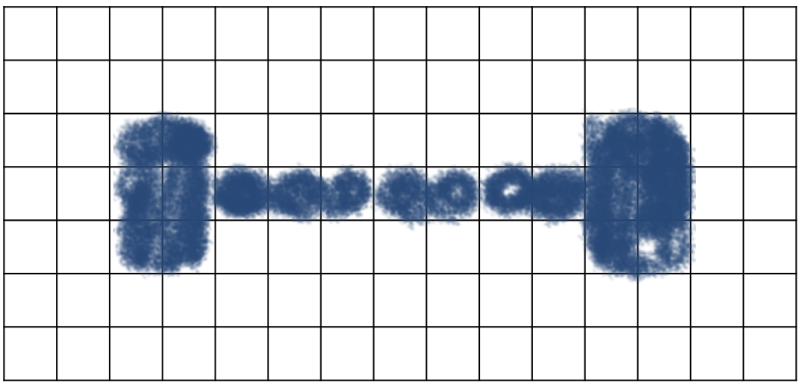
FIGURE 9.3
(a) A binary image containing one object (set), A. (b) A structuring element, B. (c) Image resulting from a morphological operation (see text).



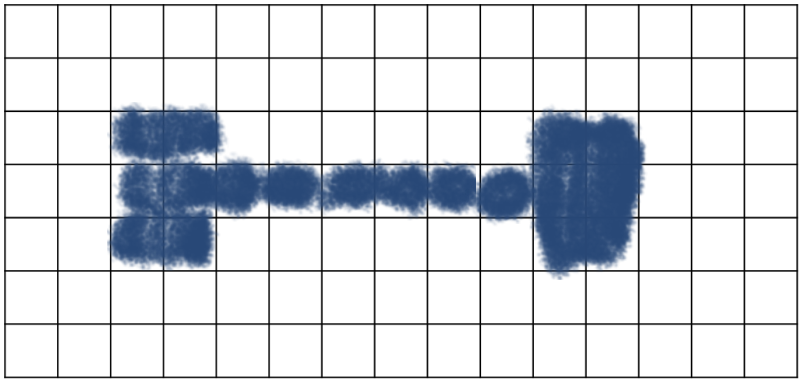
(c)



(a)



(b)



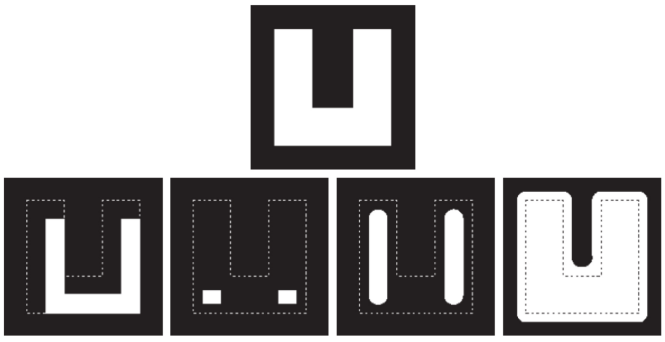
9.8 With reference to the image at the top of the figure shown below, answer the following:

(a)* Give the structuring element and morphological operation(s) that produced image (a). Show the origin of the structuring element. The dashed lines denote the boundary of the original object and are shown for reference; they are not part of the result. (The white elements are foreground pixels.)

(b) Repeat part (a) for the output shown in image (b).

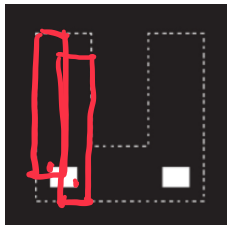
(c)* Repeat part (a) for the output shown in image (c).

(d) Repeat part (a) for the solution shown in figure (d). Note that in image (d) all corners are rounded.

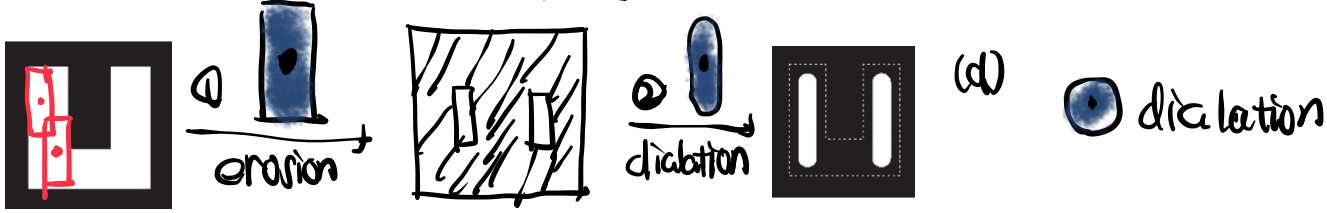


(a)* **(b)** **(c)*** **(d)**

(a) erosion (b) erosion



(c) ① erosion ② dilation



(d) dilation